ISEA INTERNSHIP

Assignment 7: Secure Network Application Development Using TCP

NAME : PIYUSH KUMAR
ROLL : CSB24051

1. Objective

The objective of this assignment is to enhance the previously developed GUI-based multi-client TCP chat application by implementing practical security mechanisms such as user authentication, password hashing, duplicate login prevention, session management, input validation, secure logging and Wireshark verification.

2. Security Features Implemented

1. User Authentication

- User enters username and password.
- Credentials are verified before login.
- Only authenticated users can access the chat room.

2. Secure Password Storage

Passwords are never stored in plain text.

Instead,

Password — > SHA-256 Hash — > Stored in users.csv

	A	B	C	D	E	F	G	H	I	J
1	username	password_hash								
2	piyush	03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4								
3	piyush_k	03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4								
4	jack	03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4								
5	marky	03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4								
6										

3. Duplicate Login Prevention

The server keeps a list of active users.

Prevents account sharing and multiple simultaneous sessions.

4. Input Validation

The application validates

- Empty username
- Empty password
- Invalid username

5. Failed Login Protection

If a user enters the wrong password repeatedly, the account get blocked temporarily.

This reduces brute-force attacks. 

3. System Architecture

Client GUI

|

Username & Password

|

TCP Connection

Authentication

|

SHA-256 Password Check

|

| |

```
Authentication Failed   Authentication Successful

    |                              |

Login Counter        Session Created

    |                              |

Block after 5 tries     Chat Communication

                    |

            Secure Logging

                    |

             Session Timeout
```

4. Implementation Steps

Step 1

Start Mininet.

sudo mn --topo single,5

Step 2

Start the server.

python3 server.py

Step 3

Run the client.

python3 client_gui.py

Step 4

Enter username and password.

Step 5

Server authenticates the user.

Step 6

If successful,

- Session starts
- User joins chat

Otherwise,

- Login rejected

Step 7

Messages are exchanged securely.

Step 8

Logout or inactivity ends the session.

5. Testing

Test Case	Expected Result	Status
Valid Login	Login Successful	Pass
Wrong Password	Login Failed	Pass
Duplicate Login	Access Denied	Pass
Empty Username	Rejected	Pass
Empty Password	Rejected	Pass
Oversized Message	Rejected	Pass
Five Wrong Attempts	Temporary Block	Pass
Logout	Session Closed	Pass
Timeout	Session Expired	Pass

6. Wireshark Verification

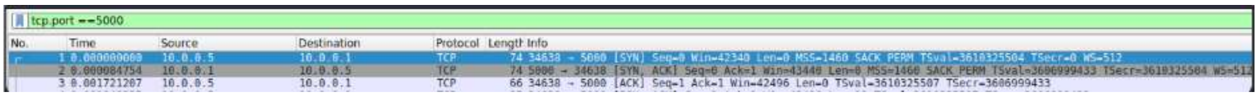
tcp.port == 5000

1. <u>Capture login</u>

   Observation :

   Login success and failure can be observed.

   TCP communication between client and server is visible.



tcp.port ==5000

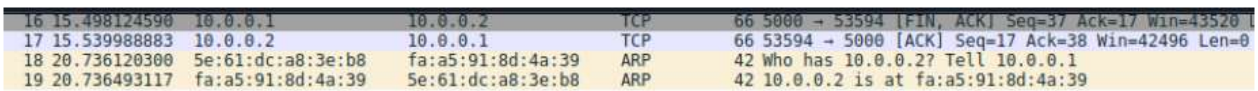
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.0.5	10.0.0.1	TCP	74	34638 → 5000 [SYN] Seq=0 Win=42340 Len=0 MSS=1460 SACK_PERM TSval=3610325504 TSecr=0 WS=512
2	0.000084754	10.0.0.1	10.0.0.5	TCP	74	5000 → 34638 [SYN, ACK] Seq=0 Ack=1 Win=43440 Len=0 MSS=1460 SACK_PERM TSval=3606999433 TSecr=3610325504 WS=512
3	0.001721207	10.0.0.5	10.0.0.1	TCP	66	34638 → 5000 [ACK] Seq=1 Ack=1 Win=42496 Len=0 TSval=3610325507 TSecr=3606999433

2. <u>Failure login</u>

   Observation :

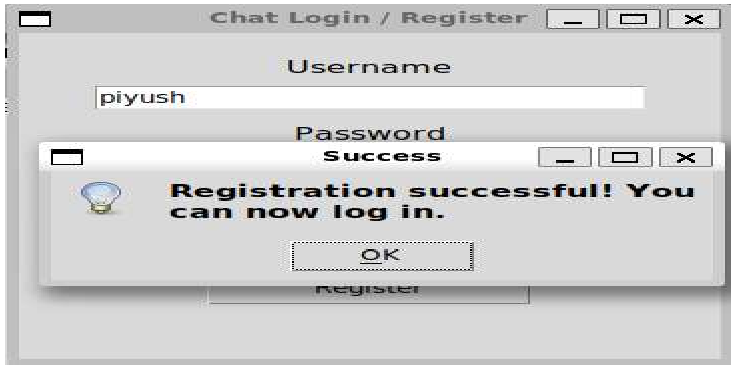
   No chat messages were exchanged after the failed authentication.

   The server did not create a user session for the client.



16	15.498124590	10.0.0.1	10.0.0.2	TCP	66	5000 → 53594 [FIN, ACK] Seq=37 Ack=17 Win=43520
17	15.539988883	10.0.0.2	10.0.0.1	TCP	66	53594 → 5000 [ACK] Seq=17 Ack=38 Win=42496 Len=0
18	20.736120300	5e:61:dc:a8:3e:b8	fa:a5:91:8d:4a:39	ARP	42	Who has 10.0.0.2? Tell 10.0.0.1
19	20.736493117	fa:a5:91:8d:4a:39	5e:61:dc:a8:3e:b8	ARP	42	10.0.0.2 is at fa:a5:91:8d:4a:39

3. <u>Authenticated communication</u>
   - The TCP three-way handshake was completed successfully between the client and the server.
   - The client sent valid login credentials to the server.

- The server verified the credentials and returned a **Login Successful** response. 

7. Conclusion

In this assignment, the existing TCP-based GUI chat application was enhanced with practical security mechanisms including authentication, SHA-256 password hashing, duplicate login prevention, input validation, failed login protection, session management and secure logging. The application was tested using Mininet and verified through Wireshark, demonstrating secure and reliable client-server communication. These enhancements significantly improve the security of the application while meeting the assignment requirements.